



Predicting Severe Asthma Exacerbations

A Genome-Wide Association Study using **Random Forests** Classifiers

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Self Introduction

Mousheng Xu, PhD

Head of Biosciences, A Private Company In Singapore

- **20+ Years Experience:** Bridging computational biology, statistical genetics, and machine learning.
- **Academic Foundation:**
 - B.S. in Genetics (Wuhan University)
 - M.S. in Biochemistry & Molecular Biology, M.S. in Computer Science (University of Kansas)
 - PhD in Bioinformatics (Boston University; Harvard Medical School)
 - Postdoc in Cancer Genetic Epidemiology (Harvard T.H. Chan School of Public Health).

Key Achievements:

- - Led large-scale GWAS & NGS pipelines.
 - Founded Hubei Vidagen (Genetic Testing).
 - Scientific Breakthrough Award (ATS 2009).

The Clinical Challenge: Severe Asthma Exacerbations



Definition

Severe Asthma Exacerbations: **emergency room visit or a hospitalization** for asthma symptoms.

Prognostic Significance

The ability to predict severe asthma exacerbations would therefore have **direct prognostic significance** and might form the basis for the development of **earlier therapeutic interventions**.

The "Small N, Large P" Problem



Sample Size (N)

127 cases

290 controls

417 total



Standard Method

**Logistic
Regression**

Struggles when predictors (P)
vastly outnumber samples (N).
Cannot easily model complex
interactions.



Predictors (P)

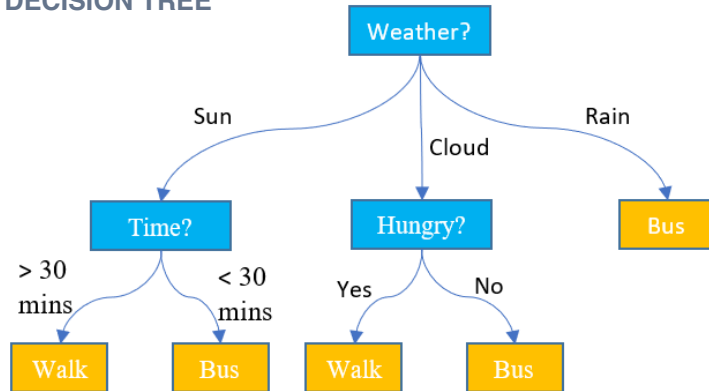
> 550,000

Single Nucleotide Polymorphisms
(SNPs) + Clinical Traits

Machine Learning With Random Forests

Leveraging ensemble power for robust predictive modeling.

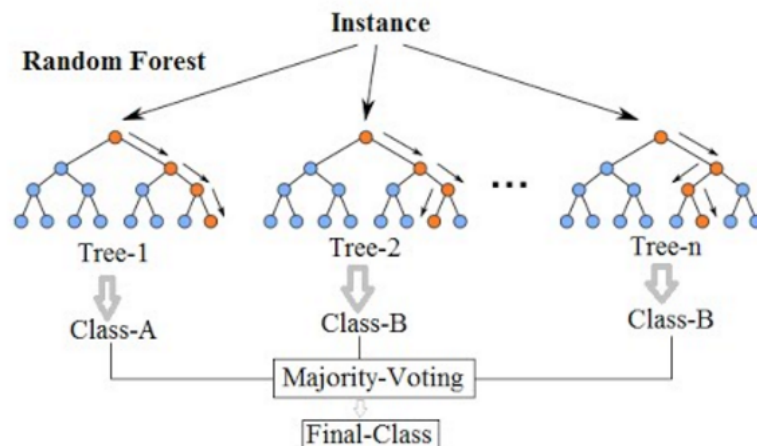
DECISION TREE



Key Concepts of Random Forests

- ✓ A **decision tree** predicts outcomes by splitting data based on feature values.
- ✓ An **ensemble of many decision trees** forms a "forest".
- ✓ Each tree uses **random samples** of the data and considers only **random features** at each split, enhancing diversity.
- ✓ **Voting Mechanism:** The model aggregates predictions by taking the **votes from each decision tree** to determine the final outcome.
- ✓ Handles **nonlinear, high-dimensional data** effectively, crucial for complex datasets.
- ✓ Provides valuable **feature importance** scores, indicating which variables are most predictive.

Random Forest Simplified

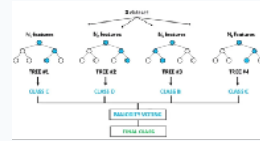


Key Terminology



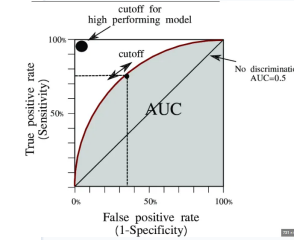
SNP

- **Single Nucleotide Polymorphism**
- A one-letter difference (A/C/G/T) on DNA
- May influence traits or disease risks
- A DNA “mutation” (inappropriate but close)



Importance Score

- Random Forest **predictive power**: contribution to the predictive accuracy
- The higher, the more contribution

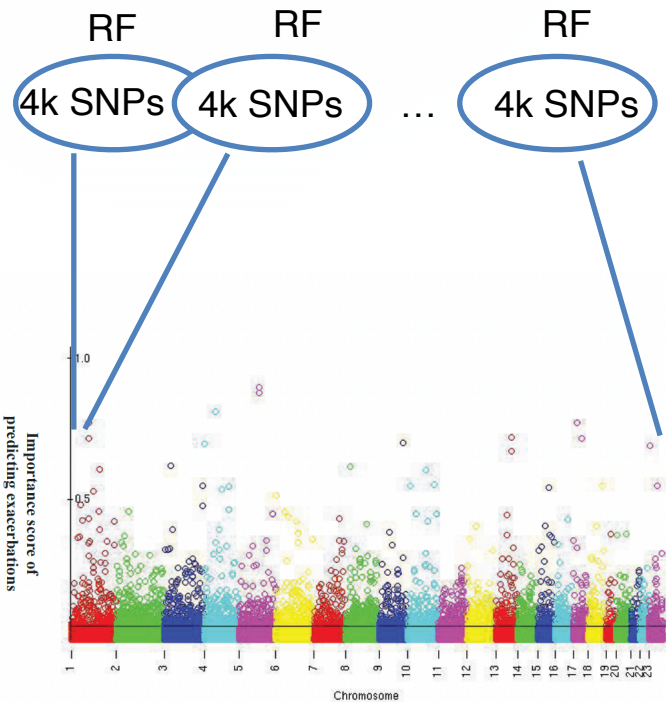


AUC

- **Area Under the Curve**
- A performance metric ranging from 0 to 1.
- Measures the ability to distinguish between classes.
- **0.5 is random guessing**
- **1.0 is perfect prediction.**

Methodology: Feature (SNP) Selection

Step 1: Initial Screen



Step 2: Ranking

Top 4k SNPs
Importance
Score



RF

Reranked 4k SNPs
Importance
Score



Step 3: Prediction

RF

Build final models using
Top N SNPs (10, 40, 160,
320) + 4 Clinical Traits.

Figure 1 The “manhattan plot” of RF importance scores of all the 550k SNPs. X-axis: the SNPs in chromosomal order; Y-axis: the RF importance scores. The black demarcation separates the top 4k SNPs from the rest.

Visualizing the Signal

The RF Importance Landscape

Unlike a traditional GWAS Manhattan plot which shows $-\log(p\text{-values})$, this plot visualizes **Random Forest Importance Scores** across the genome.

The "peaks" represent SNPs that contribute most significantly to classifying a patient as an "Exacerbator" vs "Non-Exacerbator".

We identified a clear signal distinguishing the top predictive SNPs from the genomic background noise.

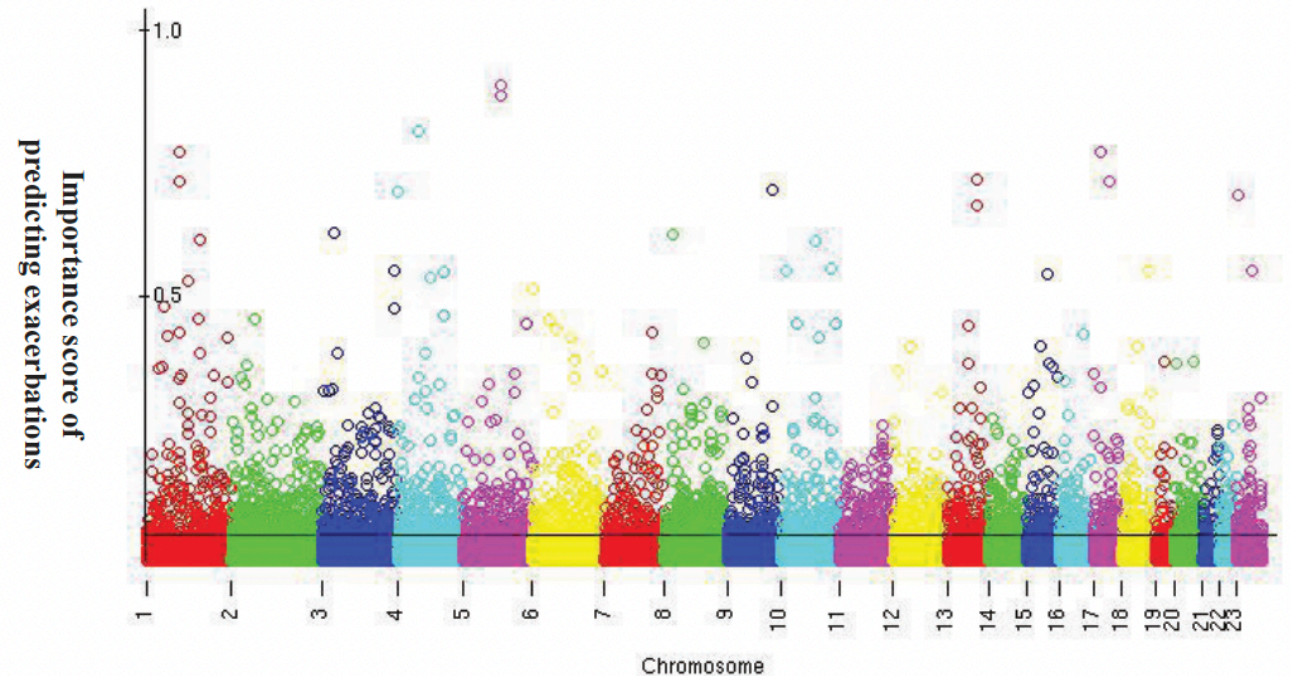
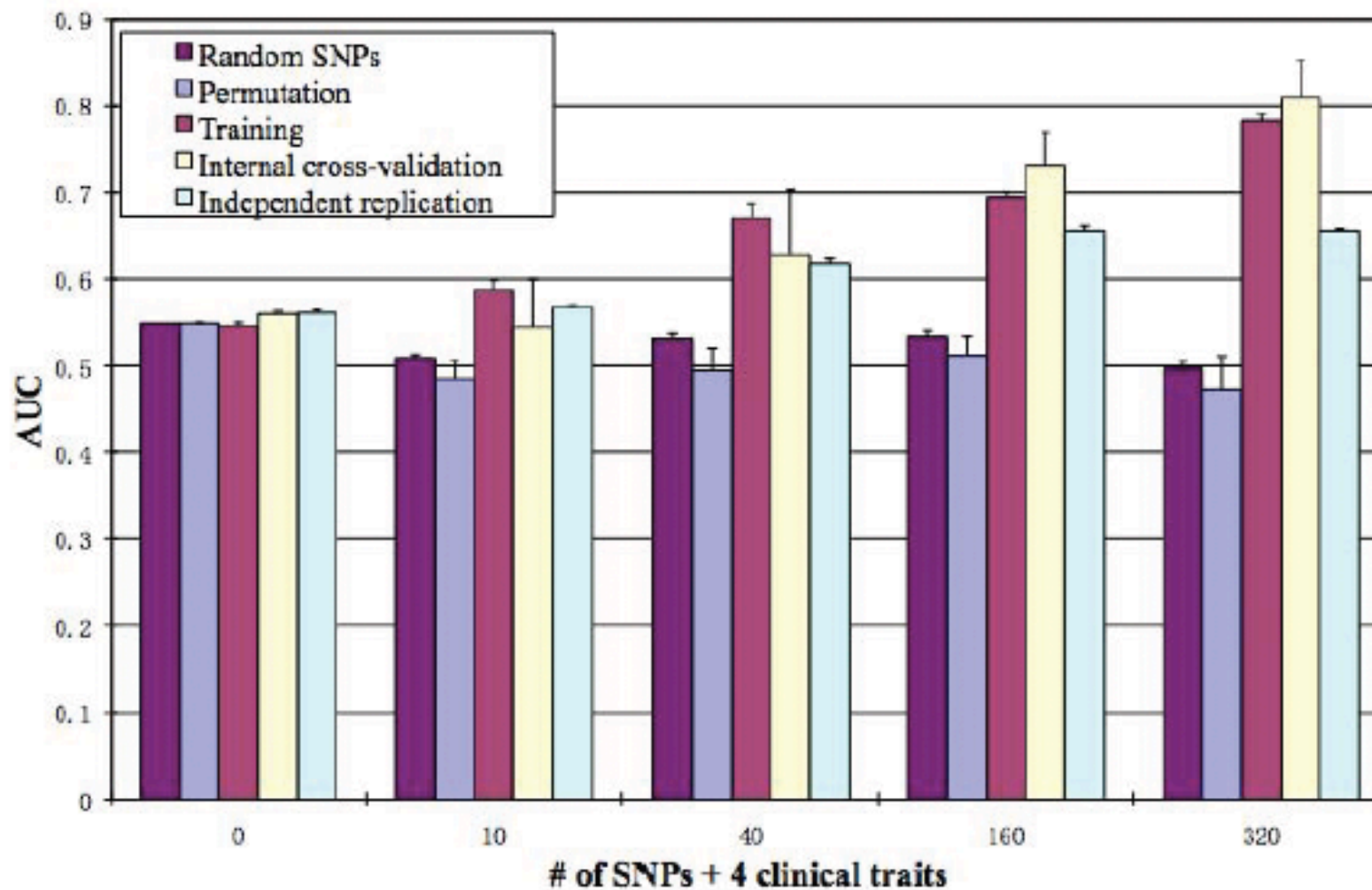


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4 months after publication, Editor:

Genetics Improves Prediction

Area Under the Curve (AUC) in Independent Replication Cohort



Key Finding: Adding genetic markers significantly improved prediction over clinical history alone. The model plateaued at 160 SNPs, suggesting a robust genetic signature for exacerbation risk.

Validation & Impact



Rigorous Validation

We did not just test on the training data.

- **Permutation Testing:** Confirmed results were not due to chance.
- **Independent Replication:** The model was tested on a separate cohort of 164 subjects, maintaining an AUC of 0.66.



Recognition

This work was highlighted for its innovative approach:

- **ATS 2009:** Featured at the "Scientific Breakthroughs of the Year" symposium.
- **Impact:** Categorized as "Highly Accessed" (1,120 accesses 4 months after publication) by BMC Medical Genetics, influencing future ML-GWAS integration studies.

Selected Publications: GWAS, NGS, Functional Genomics

GWAS

- **Handedness GWAS (Meta-analysis)** — Nat Hum Behav (2020)
- **Basal Cell Carcinoma** — PLoS One (2011)
- **Nevus Count / Melanoma Risk (NID1)** — Hum Mol Genet (2011, Co-first)
- **Cutaneous BCC & SCC** — Hum Mol Genet (2011)
- **Pancreatic Adenocarcinoma Survival** — Gut (2012)
- **Severe Asthma Exacerbation (RF GWAS)** — BMC Med Genet (2011)
- **eQTL Mapping** — Hum Mol Genet (2010)
- **ABO & Pancreatic Cancer** — Cancer Epidemiol Biomarkers Prev (2010)
- **Blood Type & Breast Cancer** — Int J Cancer (2011)

NGS & Functional Genomics

- **Synthetic transcription elongation factors** — Science (2017)
- **YAP1 → USP31 → NFκB (Sarcoma)** — Cancer Research (2018)
- **TRIM24 degrader** — Nat Chem Biol (2018)
- **MYC dependency in ovarian cancer** — eLife (2018)
- **Super-enhancer BCL2 mechanism** — Cancer Cell (2018)
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Functional Genomics, ML & Industry Highlights

Functional Genomics & Machine Learning

- **LSTM Stock Prediction** — Kaggle Tutorial (2018)
- **ENCODE Pilot Project** — Nature (2007)
- **Regulatory polymorphisms / eQTL** — Genomics (2008)
- **RF model for asthma exacerbation** — BMC Med Genet (2011)
- **Warfarin pharmacogenetics** — Pharmacogenomics (2015)
- **Biological information retrieval** — ACM CIKM (1998)
- **Lgl1 regulation during alveolarization** — Pediatric Research (2010)

Industry Highlights

- **Vidagen (Founder):**
 - Genetic testing; dozens of products; a dozen patents
 - Dozens of products; ~ a dozen patents; ~ 10s hospitals
 - Awards: Hubei Hundred, Yellow Crane, 3551 Talent
- **Booz Allen Hamilton (Sr. Data Scientist):**
 - Genomic data preparation for Million Veteran Program (MVP)
 - **Natural Language Processing (NLP)** for NIDDK
- **D2M Biotherapeutics: Causal gene identification** for cancers and immune diseases.

Vision: Data Science in BFE

Integrating Computation & Biology

My goal is to establish a functional lab at GTIIT that bridges the gap between biological engineering and data science.

- **Curriculum:** Introducing practical ML and Bioinformatics training for students.
- **Research:** Developing explainable AI pipelines for complex food and biological data.
- **Innovation:** Translating large-scale "Big Data" into actionable engineering insights.



Thank You

Questions & Discussion

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